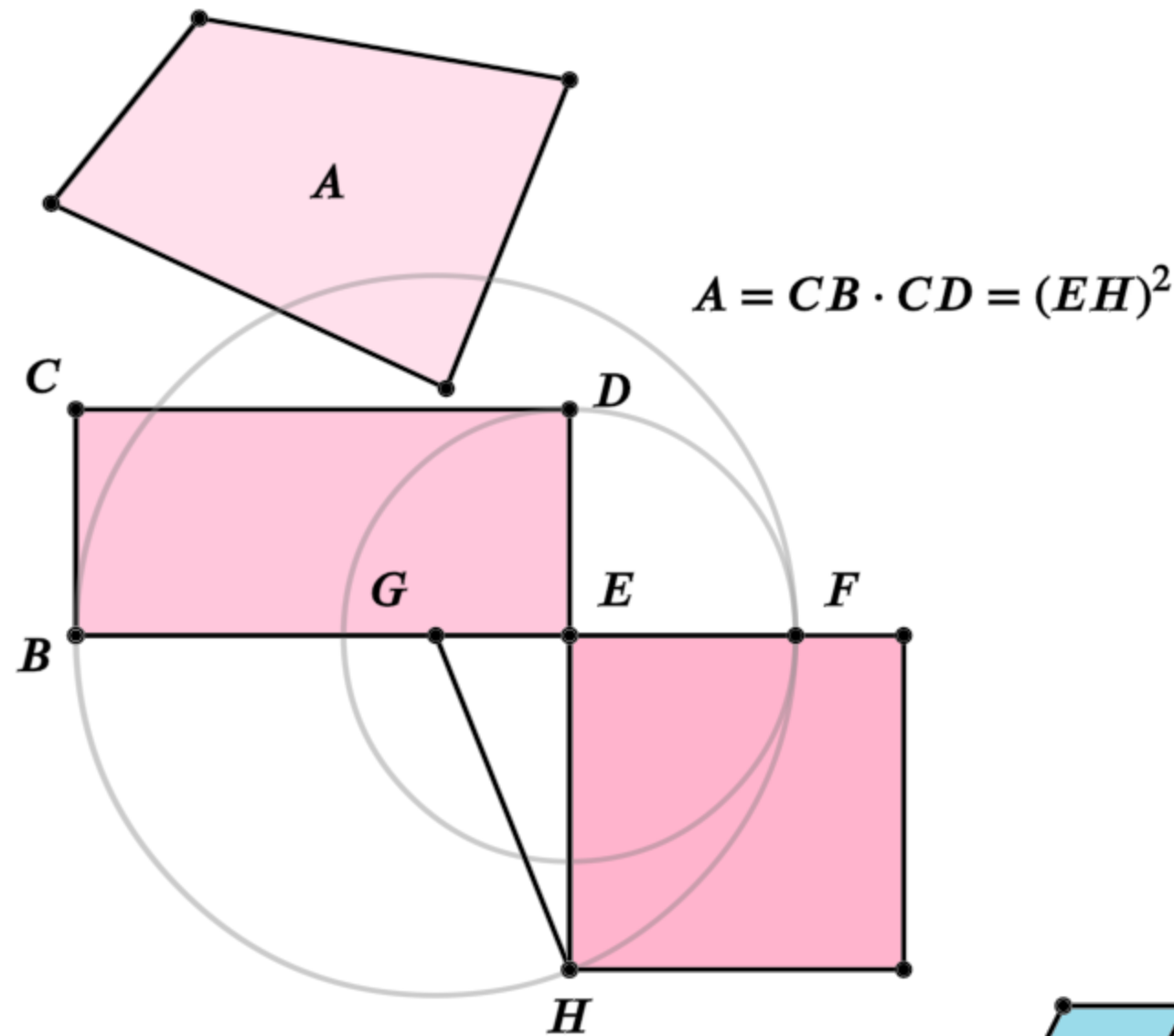


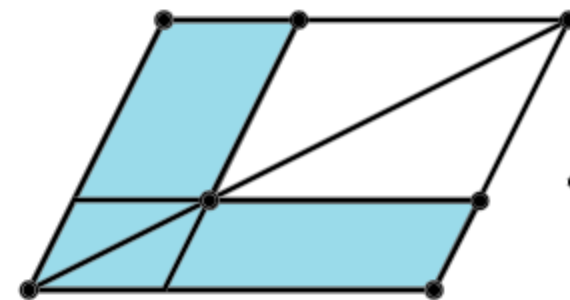
Euclid's Elements

Book II



It is a remarkable fact in the history of geometry, that the Elements of Euclid, written two thousand years ago, are still regarded by many as the best introduction to the mathematical sciences.

- Florian Cajori,
A History of Mathematics (1893)



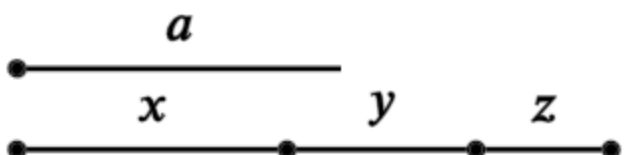
Definitions:

Any rectangular parallelogram is said to be contained by the two straight lines containing the right angle.

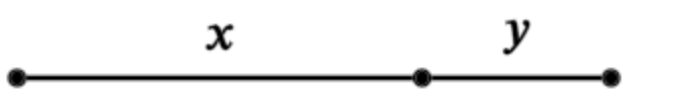
And in any parallelogrammic area let any one whatever of the parallelograms about its diameter with the two complements be called a gnomon.

Table of Contents - Book 2

1.


$$a(x + y + z) = ax + ay + az$$

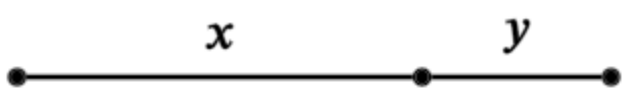
2.


$$(x + y)^2 = x(x + y) + y(x + y)$$

3.


$$y(x + y) = yx + y^2$$

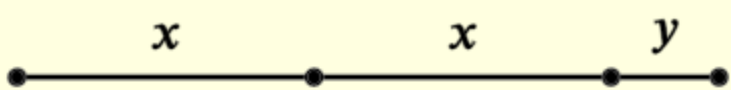
4.


$$(x + y)^2 = x^2 + y^2 + 2xy$$

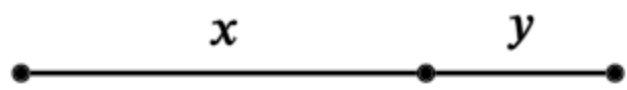
5.


$$(x + y)(x - y) + y^2 = x^2$$

6.


$$y(2x + y) + x^2 = (x + y)^2$$

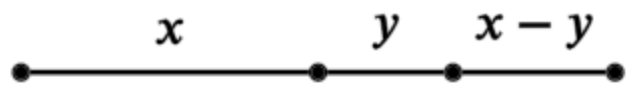
7.


$$(x + y)^2 + y^2 = x^2 + 2y(x + y)$$

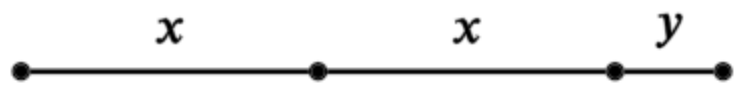
8.


$$(x + 2y)^2 = 4y(x + y) + x^2$$

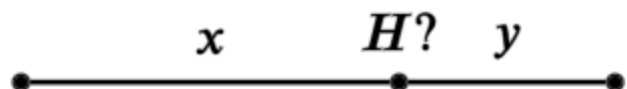
9.


$$(x + y)^2 + (x - y)^2 = 2(x^2 + y^2)$$

10.

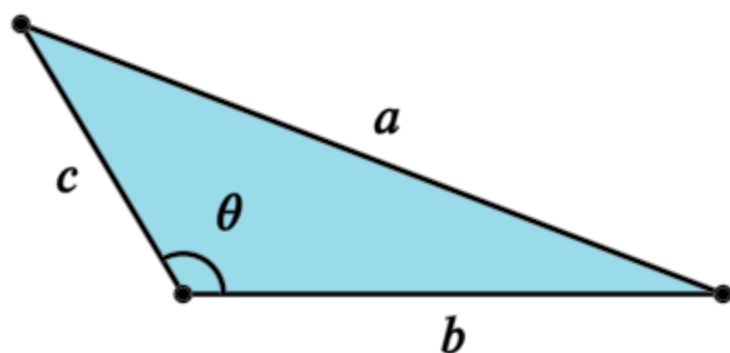

$$(2x + y)^2 + y^2 = 2(x^2 + (x + y)^2)$$

11.



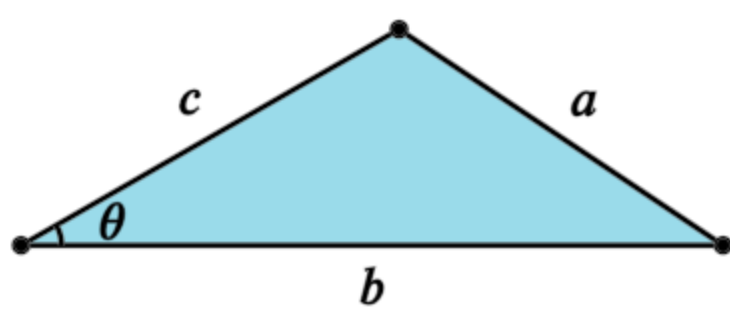
Find H such that: $y(x + y) = x^2$

12.



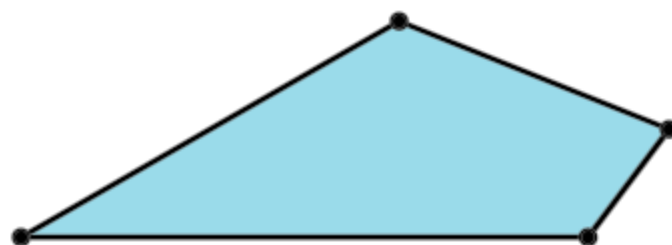
Cosine Law, $a^2 = b^2 + c^2 - 2bc \cos \theta$

13.



Cosine Law. $a^2 = b^2 + c^2 - 2bc \cos \theta$

14.



Construct a square equal to the polygon

Proposition 6 of Book 2

If a straight line be bisected and a straight line be added to it in a straight line, the rectangle contained by the whole with the added straight line and the added straight line together with the square on the half is equal to the square on the straight line made up of the half and the added straight line.



In other words

Let AB be a straight line, bisected at point C, and extend the line AB to an arbitrary point D

$$AD = AB + BD, \quad AC = CB, \quad CD = CB + BD$$

$$AD = 2x + y, \quad CD = (x + y)$$

Proposition 6 of Book 2

If a straight line be bisected and a straight line be added to it in a straight line, the rectangle contained by the whole with the added straight line and the added straight line together with the square on the half is equal to the square on the straight line made up of the half and the added straight line.



In other words

Let AB be a straight line, bisected at point C, and extend the line AB to an arbitrary point D

The rectangle formed by the extended line AD, and the extension BD plus the square on CB is equal to the square on CD

$$AD = AB + BD, \quad AC = CB, \quad CD = CB + BD$$

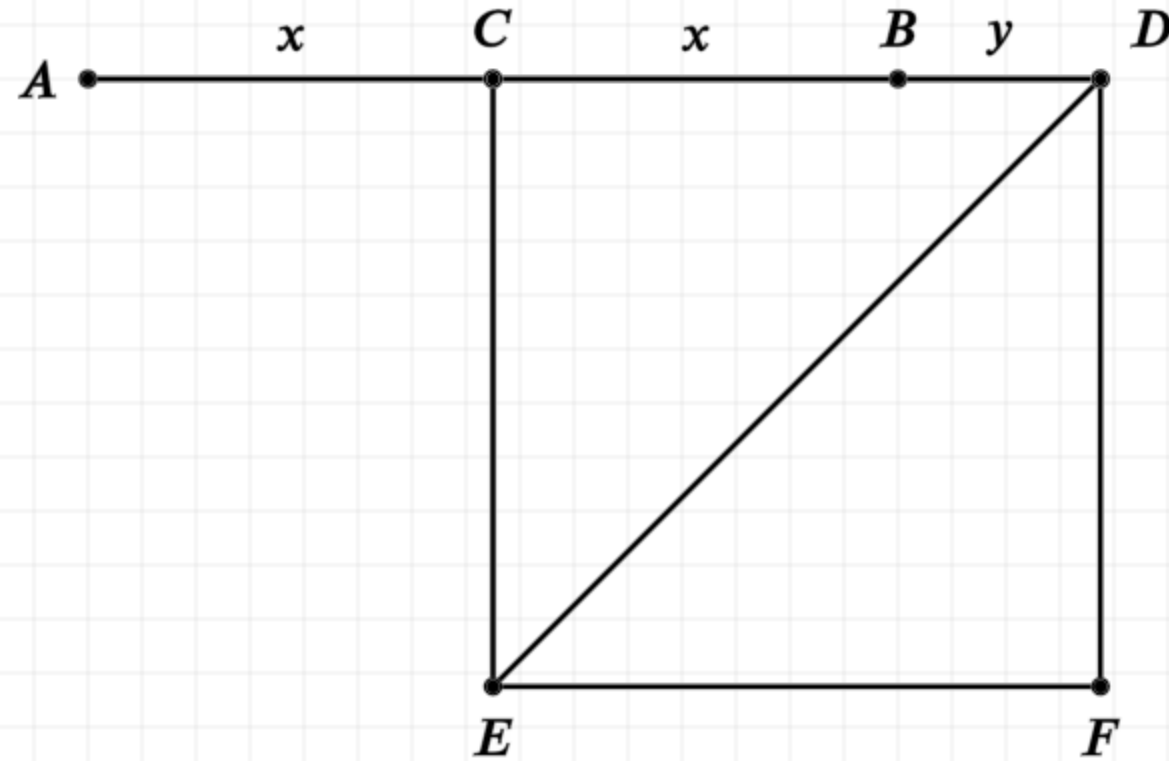
$$AD \cdot DB + CB^2 = CD^2$$

$$AD = 2x + y, \quad CD = (x + y)$$

$$(2x + y)y + x^2 = (x + y)^2$$

Proposition 6 of Book 2

If a straight line be bisected and a straight line be added to it in a straight line, the rectangle contained by the whole with the added straight line and the added straight line together with the square on the half is equal to the square on the straight line made up of the half and the added straight line.



In other words

Let AB be a straight line, bisected at point C, and extend the line AB to an arbitrary point D

$$AD \cdot DB + CB^2 = CD^2$$

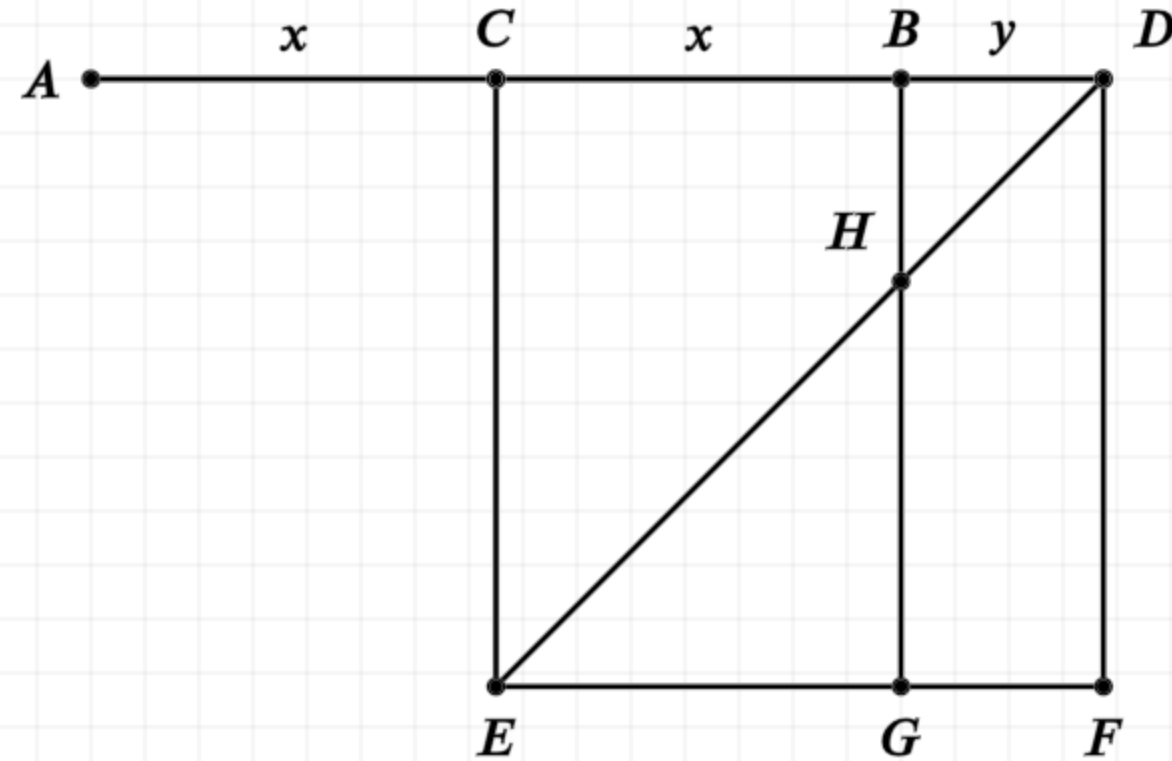
Construction:

Draw a square CEFB on the line CD (I.46) and draw the diagonal DE

$$AD = AB + BD, \quad AC = CB, \quad CD = CB + BD$$

Proposition 6 of Book 2

If a straight line be bisected and a straight line be added to it in a straight line, the rectangle contained by the whole with the added straight line and the added straight line together with the square on the half is equal to the square on the straight line made up of the half and the added straight line.



$$AD = AB + BD, \quad AC = CB, \quad CD = CB + BD$$

In other words

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$$AD \cdot DB + CB^2 = CD^2$$

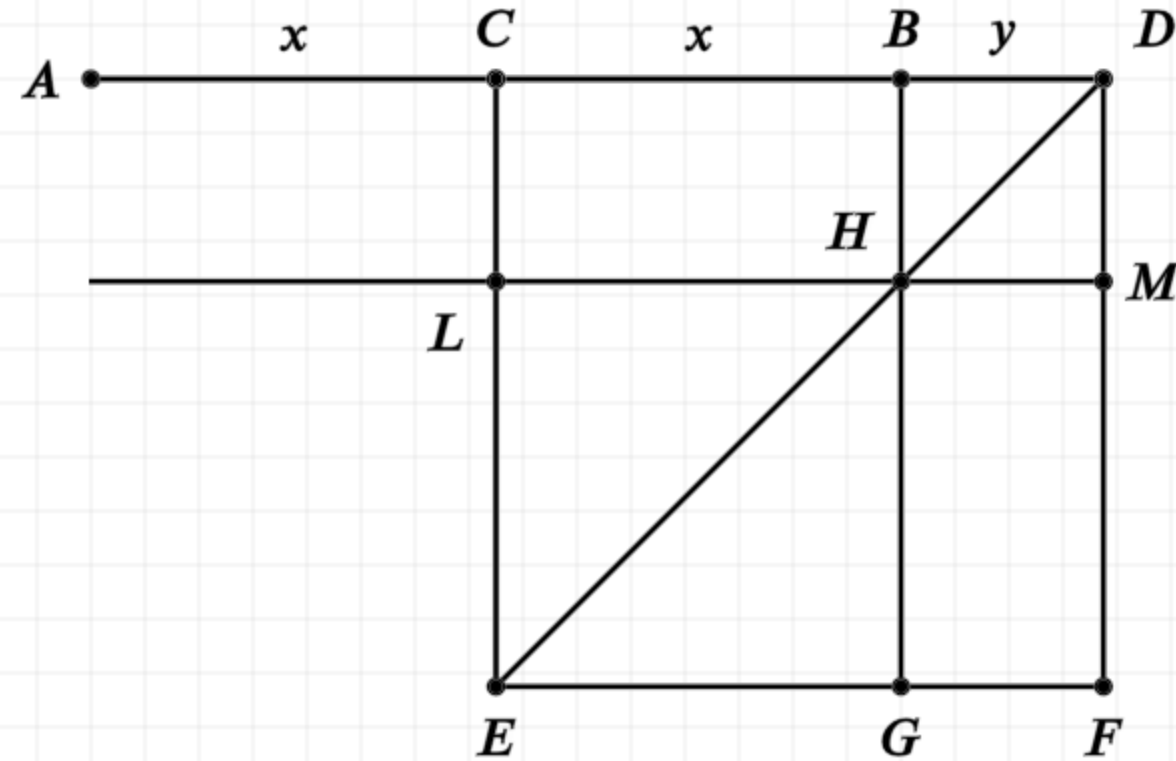
Construction:

Draw a square CEFG on the line CD (I.46) and draw the diagonal DE

From point B, draw a line parallel to either CE or DF (I.31)

Proposition 6 of Book 2

If a straight line be bisected and a straight line be added to it in a straight line, the rectangle contained by the whole with the added straight line and the added straight line together with the square on the half is equal to the square on the straight line made up of the half and the added straight line.



$$AD = AB + BD, \quad AC = CB, \quad CD = CB + BD$$

In other words

Let AB be a straight line, bisected at point C, and extend the line AB to an arbitrary point D

$$AD \cdot DB + CB^2 = CD^2$$

Construction:

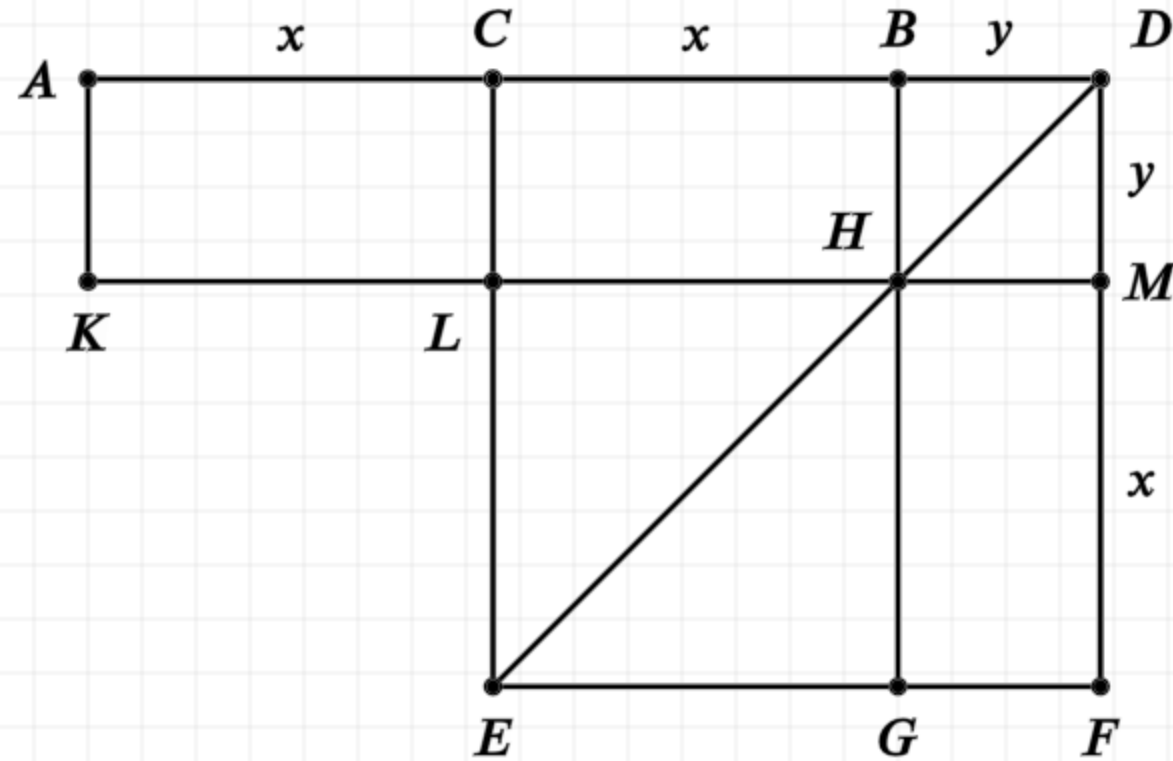
Draw a square CEFG on the line CD (I.46) and draw the diagonal DE

From point B, draw a line parallel to either CE or DF (I.31)

From point H, draw a line parallel to either AB or EF (I.31)

Proposition 6 of Book 2

If a straight line be bisected and a straight line be added to it in a straight line, the rectangle contained by the whole with the added straight line and the added straight line together with the square on the half is equal to the square on the straight line made up of the half and the added straight line.



$$AD = AB + BD, \quad AC = CB, \quad CD = CB + BD$$

In other words

Let AB be a straight line, bisected at point C, and extend the line AB to an arbitrary point D

$$AD \cdot DB + CB^2 = CD^2$$

Construction:

Draw a square CEFB on the line CD (I.46) and draw the diagonal DE

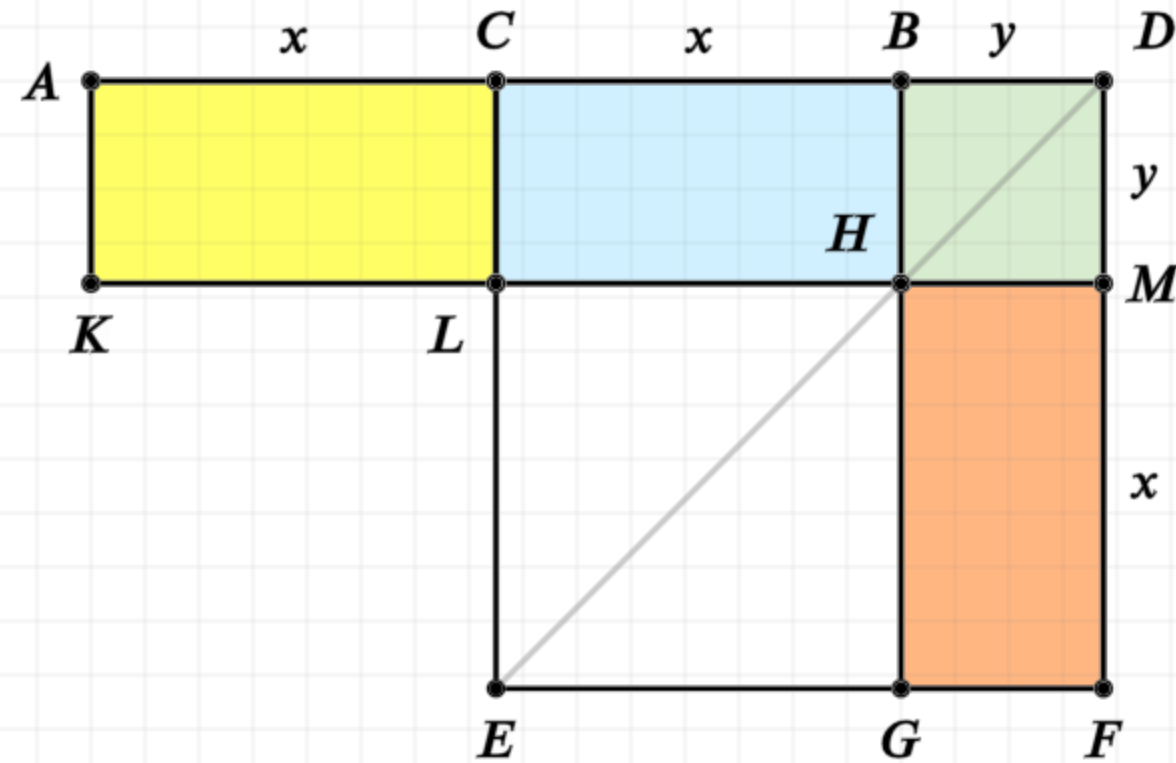
From point B, draw a line parallel to either CE or DF (I.31)

From point H, draw a line parallel to either AB or EF (I.31)

From point A, draw a line parallel to either CL or BH (I.31)

Proposition 6 of Book 2

If a straight line be bisected and a straight line be added to it in a straight line, the rectangle contained by the whole with the added straight line and the added straight line together with the square on the half is equal to the square on the straight line made up of the half and the added straight line.



In other words

Let AB be a straight line, bisected at point C, and extend the line AB to an arbitrary point D

$$AD \cdot DB + CB^2 = CD^2$$

Proof:

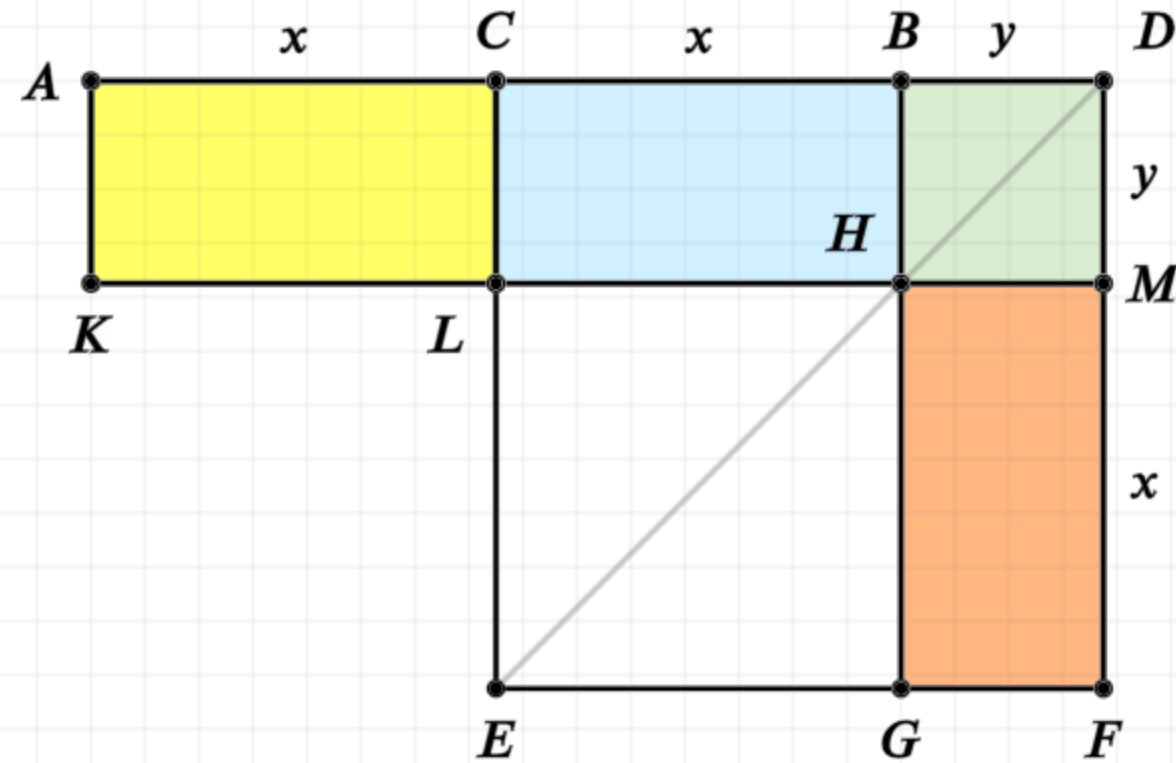
Since AC equals CB, then AL equals CH (I.36), and CH equals HF (I.43), then $AL = HF$

$$AD = AB + BD, \quad AC = CB, \quad CD = CB + BD$$

$$\text{Yellow } AL = \text{Light Blue } CH = \text{Orange } HF$$

Proposition 6 of Book 2

If a straight line be bisected and a straight line be added to it in a straight line, the rectangle contained by the whole with the added straight line and the added straight line together with the square on the half is equal to the square on the straight line made up of the half and the added straight line.



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Let AB be a straight line, bisected at point C, and extend the line AB to an arbitrary point D

$$AD \cdot DB + CB^2 = CD^2$$

Proof:

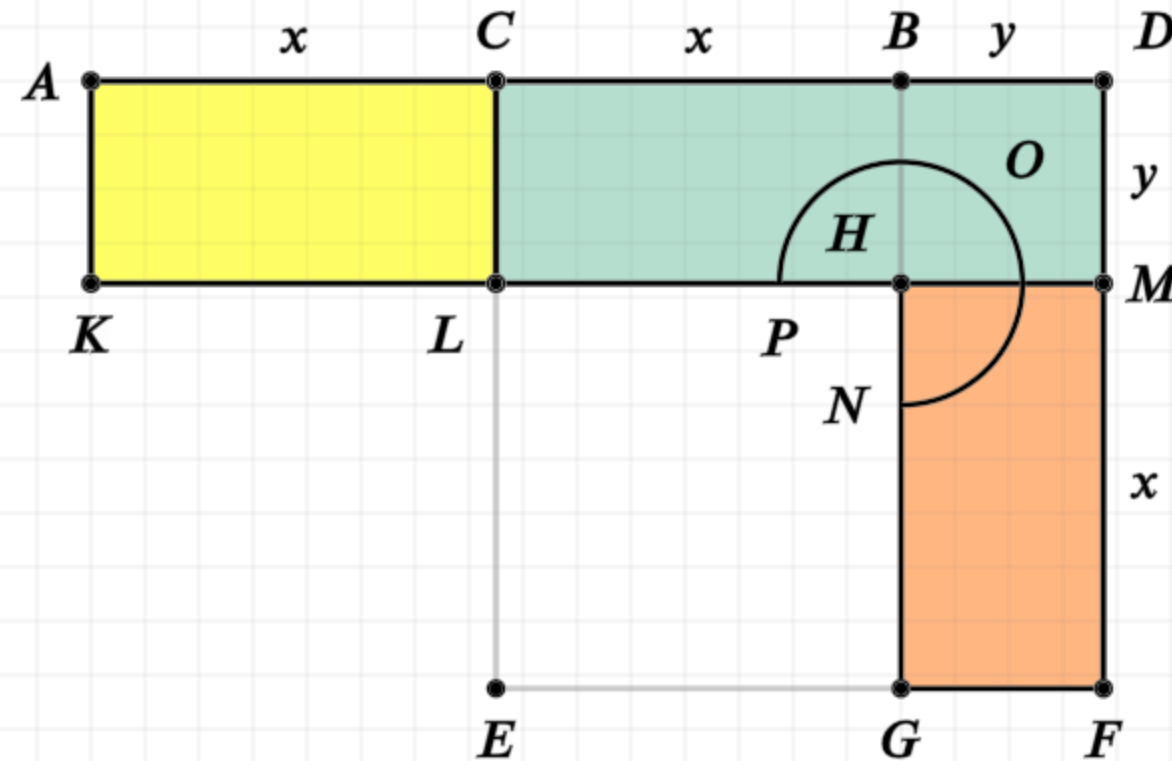
Since AC equals CB, then AL equals CH (I.36), and CH equals HF (I.43), then AL = HF

$$AD = AB + BD, \quad AC = CB, \quad CD = CB + BD$$

$$\text{Yellow } AL = \text{Light Blue } CH = \text{Orange } HF, \quad \text{Yellow } AL = \text{Orange } HF$$

Proposition 6 of Book 2

If a straight line be bisected and a straight line be added to it in a straight line, the rectangle contained by the whole with the added straight line and the added straight line together with the square on the half is equal to the square on the straight line made up of the half and the added straight line.



In other words

Let AB be a straight line, bisected at point C, and extend the line AB to an arbitrary point D

$$AD \cdot DB + CB^2 = CD^2$$

Proof:

Since AC equals CB, then AL equals CH (I.36), and CH equals HF (I.43), then $AL = HF$

Add CM to both AL and HF

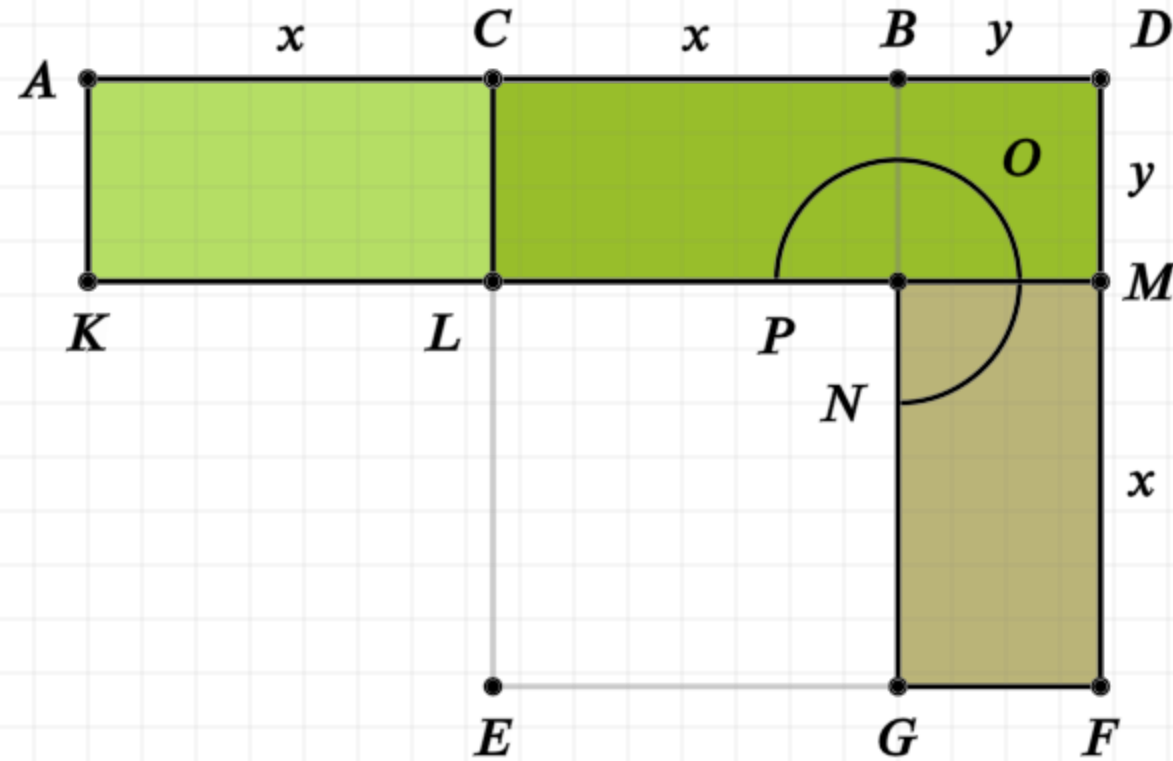
$$AD = AB + BD, \quad AC = CB, \quad CD = CB + BD$$

$$\text{Yellow } AL = \text{Light Blue } CH = \text{Orange } HF, \quad \text{Yellow } AL = \text{Orange } HF$$

$$\text{Yellow } AL + \text{Green } CM = \text{Orange } HF + \text{Green } CM$$

Proposition 6 of Book 2

If a straight line be bisected and a straight line be added to it in a straight line, the rectangle contained by the whole with the added straight line and the added straight line together with the square on the half is equal to the square on the straight line made up of the half and the added straight line.



In other words

Let AB be a straight line, bisected at point C, and extend the line AB to an arbitrary point D

$$AD \cdot DB + CB^2 = CD^2$$

Proof:

Since AC equals CB, then AL equals CH (I.36), and CH equals HF (I.43), then $AL = HF$

Add CM to both AL and HF

$$AD = AB + BD, \quad AC = CB, \quad CD = CB + BD$$

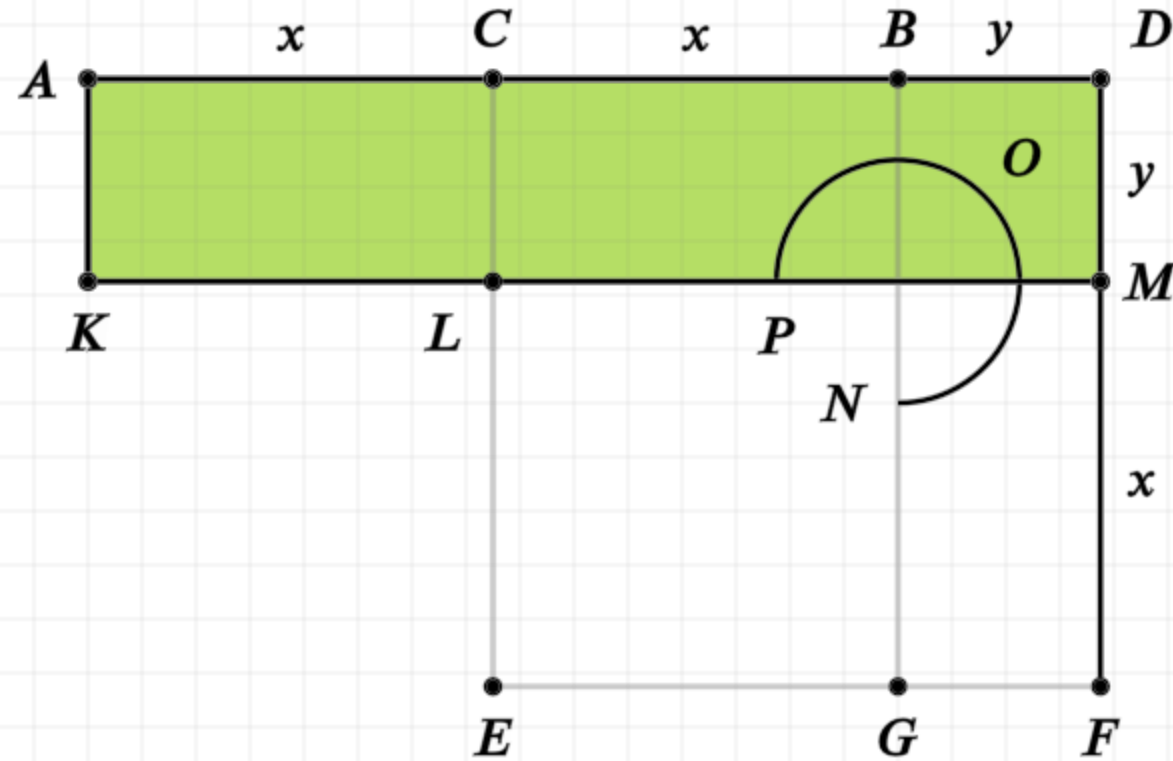
$$\text{yellow } AL = \text{light blue } CH = \text{light orange } HF, \quad \text{yellow } AL = \text{orange } HF$$

$$\text{Yellow } AL + \text{Teal } CM = \text{Orange } HF + \text{Teal } CM$$

■ $AM =$ ■ NOP

Proposition 6 of Book 2

If a straight line be bisected and a straight line be added to it in a straight line, the rectangle contained by the whole with the added straight line and the added straight line together with the square on the half is equal to the square on the straight line made up of the half and the added straight line.


$$AD = AB + BD, \quad AC = CB, \quad CD = CB + BD$$

$$\text{yellow } AL = \text{light blue } CH = \text{orange } HF, \quad \text{yellow } AL = \text{orange } HF$$

$$AL + CM = HF + CM$$

■ $AM = \blacksquare NOP$

■ $AM = AD \cdot DB$

In other words

Let AB be a straight line, bisected at point C, and extend the line AB to an arbitrary point D

$$AD \cdot DB + CB^2 = CD^2$$

Proof:

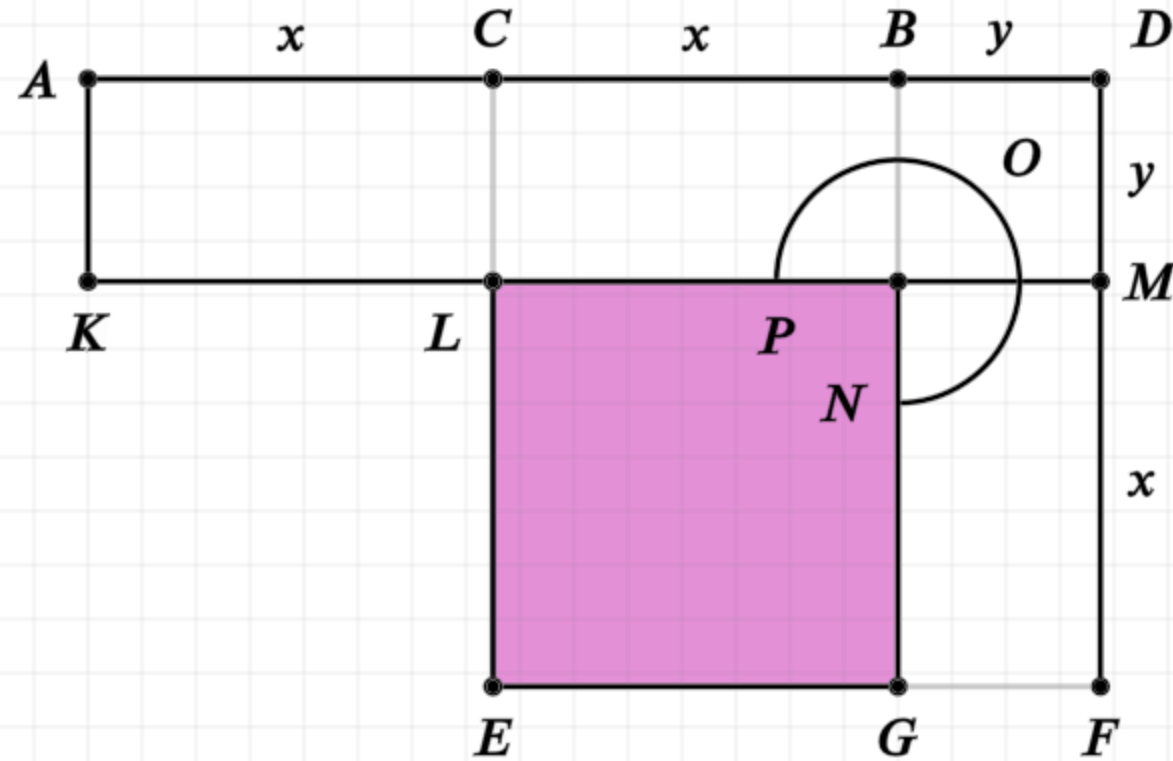
Since AC equals CB, then AL equals CH (I.36), and CH equals HF (I.43), then $AL = HF$

Add CM to both AL and HF

Since DB equals DM, AM is the rectangle formed from AD and DB

Proposition 6 of Book 2

If a straight line be bisected and a straight line be added to it in a straight line, the rectangle contained by the whole with the added straight line and the added straight line together with the square on the half is equal to the square on the straight line made up of the half and the added straight line.



$$AD = AB + BD, \quad AC = CB, \quad CD = CB + BD$$

$$AL = CH = HF, \quad AL = HF$$

$$AL + CM = HF + CM$$

$$AM = NOP$$

$$AM = AD \cdot DB$$

$$LG = CB^2$$

In other words

Let AB be a straight line, bisected at point C, and extend the line AB to an arbitrary point D

$$AD \cdot DB + CB^2 = CD^2$$

Proof:

Since AC equals CB, then AL equals CH (I.36), and CH equals HF (I.43), then $AL = HF$

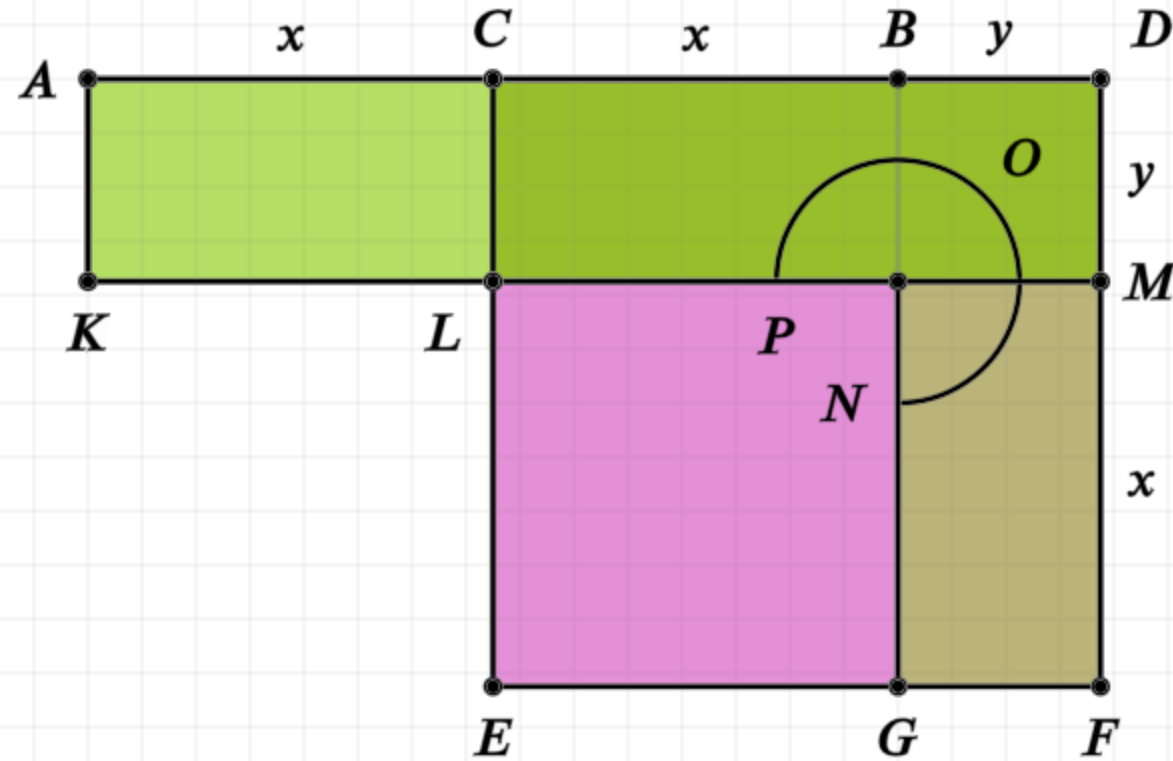
Add CM to both AL and HF

Since DB equals DM, AM is the rectangle formed from AD and DB

LG is the square of CB

Proposition 6 of Book 2

If a straight line be bisected and a straight line be added to it in a straight line, the rectangle contained by the whole with the added straight line and the added straight line together with the square on the half is equal to the square on the straight line made up of the half and the added straight line.


$$AD = AB + BD, \quad AC = CB, \quad CD = CB + BD$$
$$\text{yellow } AL = \text{light blue } CH = \text{orange } HF, \quad \text{yellow } AL = \text{orange } HF$$
$$AL + CM = HF + CM$$

■ $AM = \blacksquare NOP$

■ $AM = AD \cdot DB$

■ $LG = CB^2$

$$AD \cdot DB + CB^2 = \blacksquare NOP + CB^2$$

In other words

Let AB be a straight line, bisected at point C, and extend the line AB to an arbitrary point D

$$AD \cdot DB + CB^2 = CD^2$$

Proof:

Since AC equals CB, then AL equals CH (I.36), and CH equals HF (I.43), then $AL = HF$

Add CM to both AL and HF

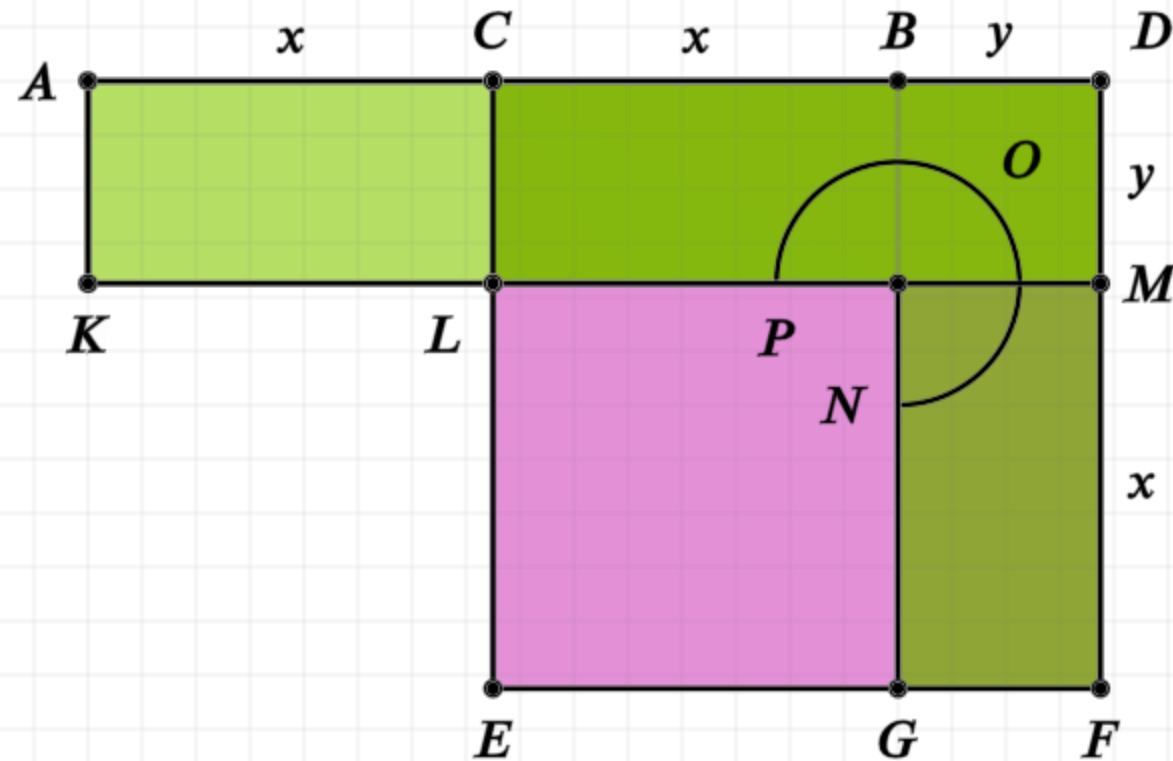
Since DB equals DM, AM is the rectangle formed from AD and DB

LG is the square of CB

Add LG to AM and the gnomon NOP

Proposition 6 of Book 2

If a straight line be bisected and a straight line be added to it in a straight line, the rectangle contained by the whole with the added straight line and the added straight line together with the square on the half is equal to the square on the straight line made up of the half and the added straight line.



In other words

Let AB be a straight line, bisected at point C, and extend the line AB to an arbitrary point D

$$AD \cdot DB + CB^2 = CD^2$$

Proof:

Since AC equals CB, then AL equals CH (I.36), and CH equals HF (I.43), then $AL = HF$

Add CM to both AL and HF

Since DB equals DM, AM is the rectangle formed from AD and DB

LG is the square of CB

Add LG to AM and the gnomon NOP

But LG added to the gnomon NOP is also equal to the square on CD, therefore AM plus LG is equal to CD^2

$$AD = AB + BD, \quad AC = CB, \quad CD = CB + BD$$

$$AL = CH = HF, \quad AL = HF$$

$$AL + CM = HF + CM$$

$$AM = NOP$$

$$AM = AD \cdot DB$$

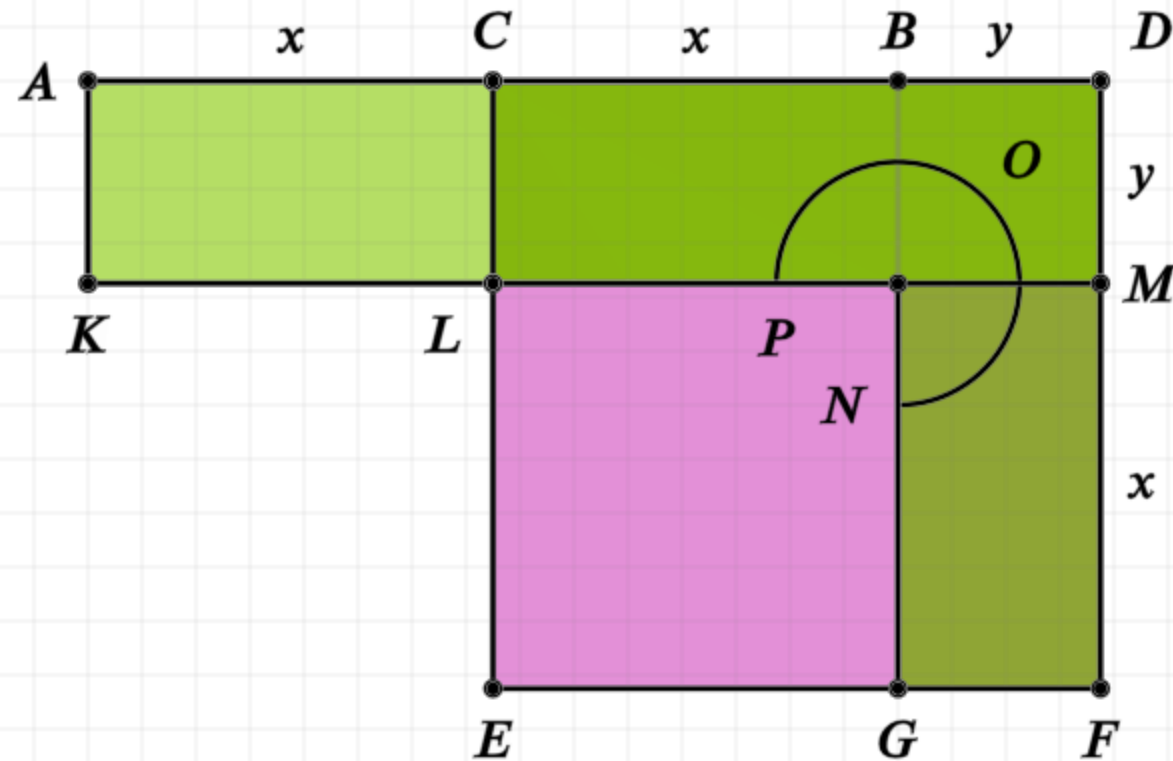
$$LG = CB^2$$

$$AD \cdot DB + CB^2 = NOP + CB^2$$

$$AD \cdot DB + CB^2 = CD^2$$

Proposition 6 of Book 2

If a straight line be bisected and a straight line be added to it in a straight line, the rectangle contained by the whole with the added straight line and the added straight line together with the square on the half is equal to the square on the straight line made up of the half and the added straight line.



$$AD = AB + BD, \quad AC = CB, \quad CD = CB + BD$$

$$AL = CH = HF, \quad AL = HF$$

$$AL + CM = HF + CM$$

$$AM = NOP$$

$$AM = AD \cdot DB$$

$$LG = CB^2$$

$$AD \cdot DB + CB^2 = NOP + CB^2$$

$$AD \cdot DB + CB^2 = CD^2$$

In other words

Let AB be a straight line, bisected at point C, and extend the line AB to an arbitrary point D

$$AD \cdot DB + CB^2 = CD^2$$

Proof:

Since AC equals CB, then AL equals CH (I.36), and CH equals HF (I.43), then $AL = HF$

Add CM to both AL and HF

Since DB equals DM, AM is the rectangle formed from AD and DB

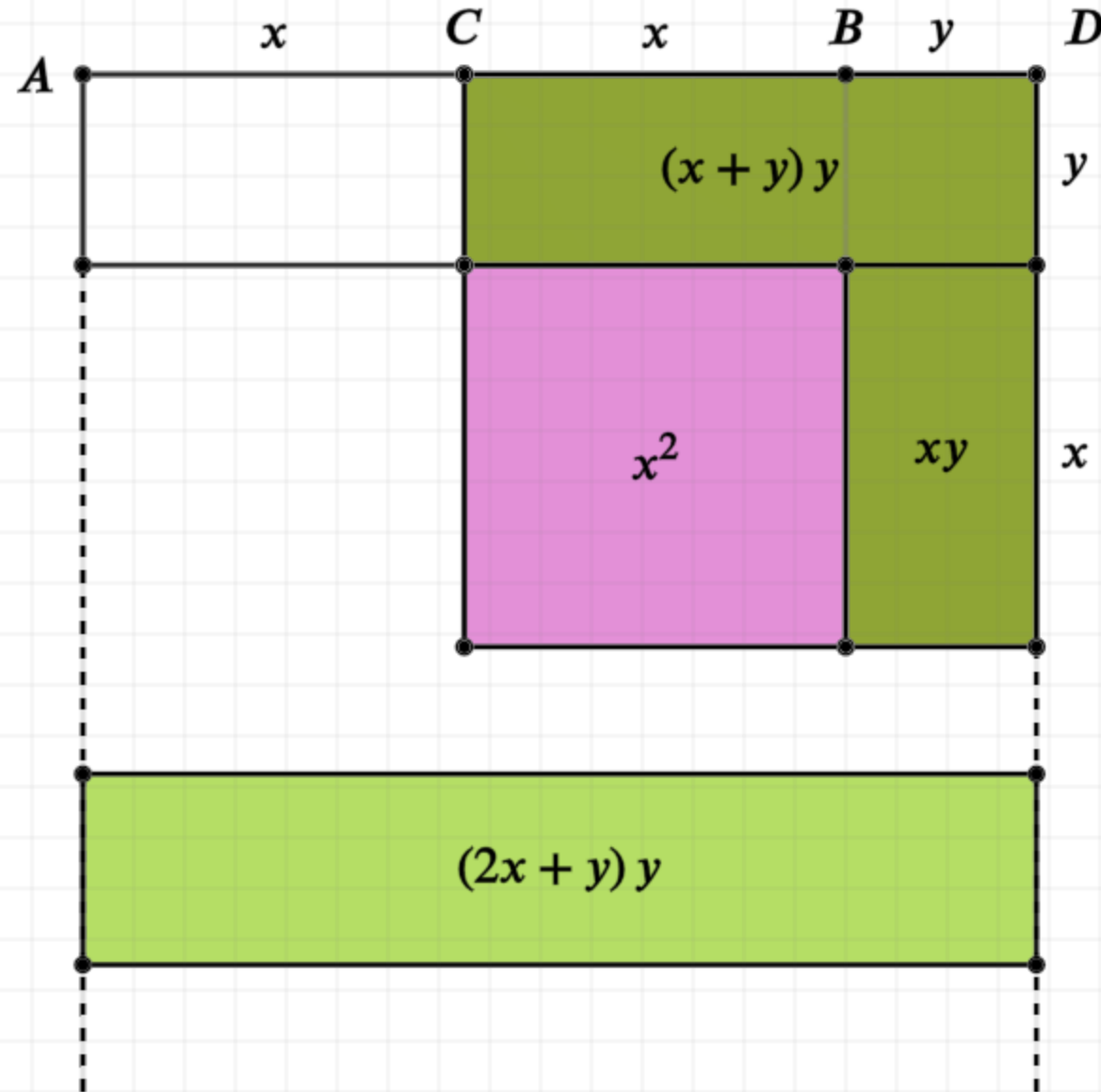
LG is the square of CB

Add LG to AM and the gnomon NOP

But LG added to the gnomon NOP is also equal to the square on CD, therefore AM plus LG is equal to CD^2

Proposition 6 of Book 2

If a straight line be bisected and a straight line be added to it in a straight line, the rectangle contained by the whole with the added straight line and the added straight line together with the square on the half is equal to the square on the straight line made up of the half and the added straight line.



$$AD \cdot DB + CB^2 = CD^2$$

$$(2x + y)y + x^2 = (x + y)^2$$

In other words

Let AB be a straight line, bisected at point C, and extend the line AB to an arbitrary point D

$$AD \cdot DB + CB^2 = CD^2$$

Proof:

Since AC equals CB, then AL equals CH (I.36), and CH equals HF (I.43), then AL = HF

Add CM to both AL and HF

Since DB equals DM, AM is the rectangle formed from AD and DB

LG is the square of CB

Add LG to AM and the gnomon NOP

But LG added to the gnomon NOP is also equal to the square on CD, therefore AM plus LG is equal to CF

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